

Cascode Current Mirror :

The voltage at point R is $V_{GS3} + V_{GS4}$. Hence voltage at point Q is $V_{GS3} + V_{GS4} - V_{GS2}$. Hence minimum voltage at point P is $V_{GS3} + V_{GS4} - V_{GS2} + V_{ov2}$. If all V_{GS} values are equal, $V_p(min) = V_{GS} + V_{ov} = 2V_{ov} + V_{thn}$.

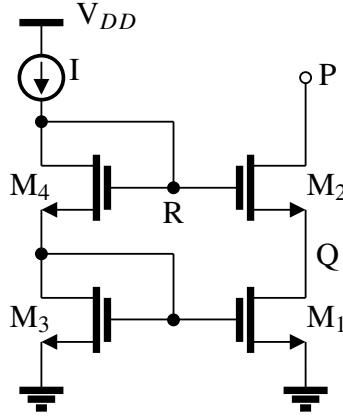


Figure 1: Cascode current mirror

Hence voltage headroom is high across P i.e. output of this mirror.

Problem : Design cascode current mirror with minimum voltage headroom at point P of 0.6 V. Neglect body effect and assume that size of all transistors is same. Current flowing through all MOSFETs is $40 \mu A$. Given $\mu_n C_{ox} = 300 \mu A/V^2$, $V_{thn} = 0.38 V$.

$$V_p(min) = 2V_{ov} + V_{thn}$$

$$0.6 = 2V_{ov} + 0.38$$

$$V_{ov} = 0.11V$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{ov}^2$$

$$40 \times 10^{-6} = \frac{1}{2} \times 300 \times 10^{-6} \times \frac{W}{L} \times (0.11)^2$$

$$\frac{W}{L} = 22.04$$

Low Voltage Cascode Current Mirror : In order to minimize voltage headroom at point P, $V_b = V_{ov2} + V_{GS3}$. But in this case $V_{DS2} \neq V_{DS1} = V_{GS1}$ and current copying will not be exact. This is due to diode connection of M_1 .

We can force V_{DS1} to be equal to V_{DS2} with following modified circuit.

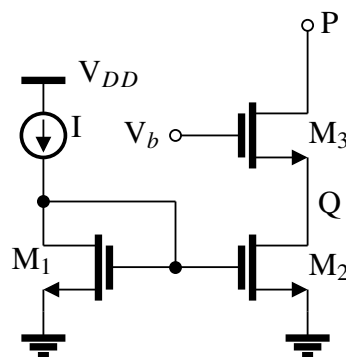
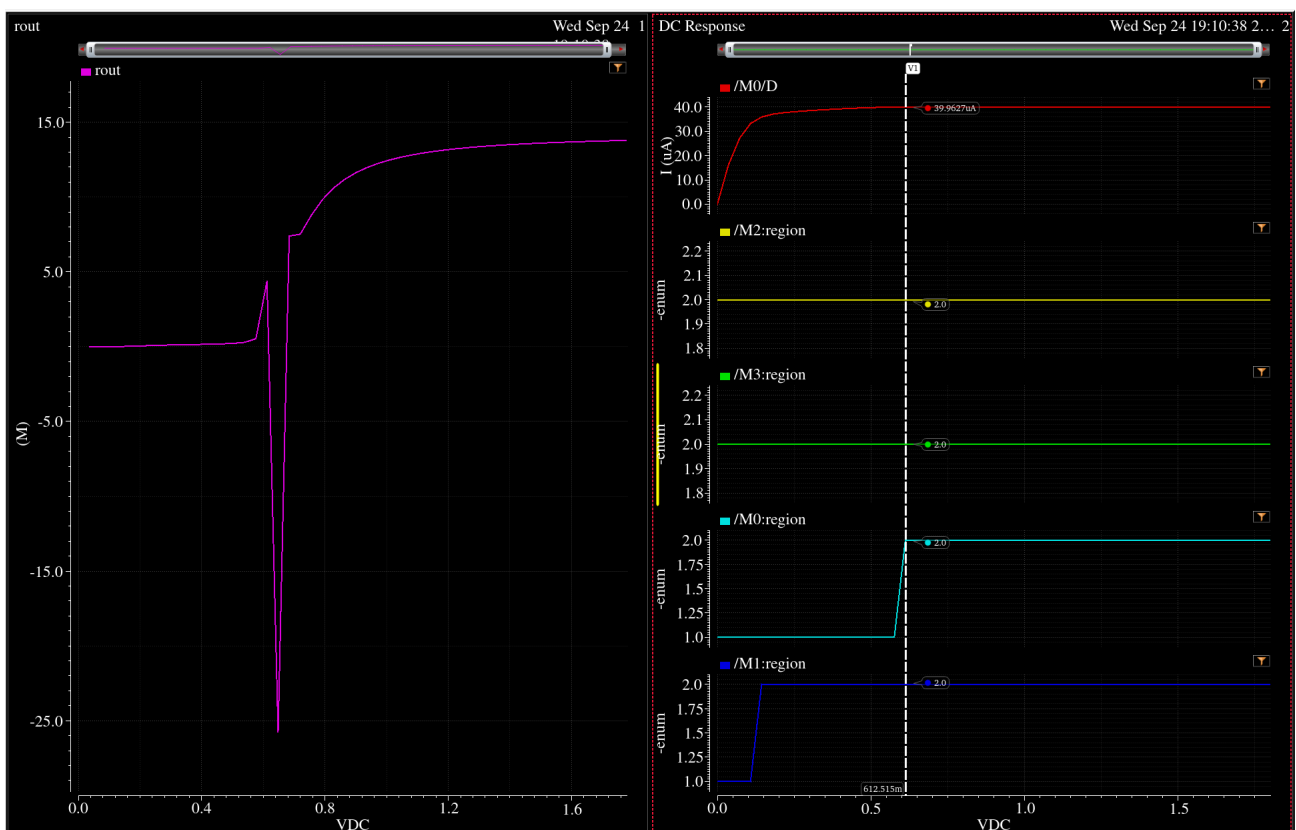
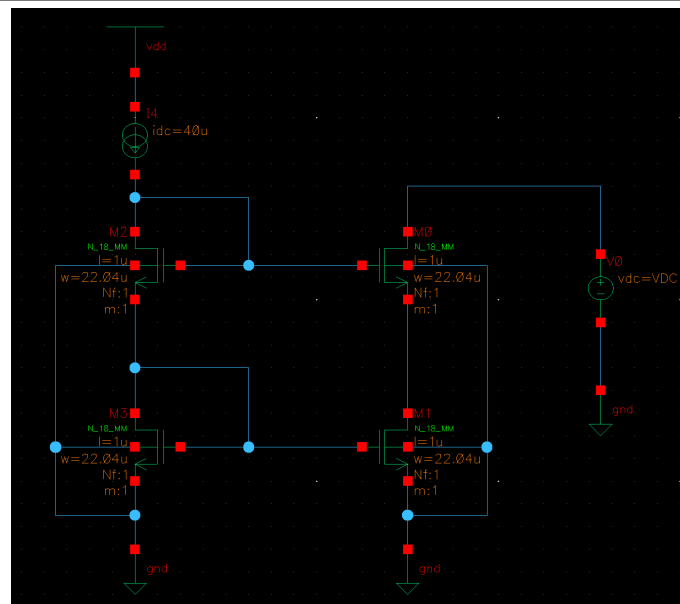


Figure 2: Low voltage cascode current mirror

For M_1 to be in saturation

$$V_{DS1} \geq V_{GS1} - V_{th1}$$

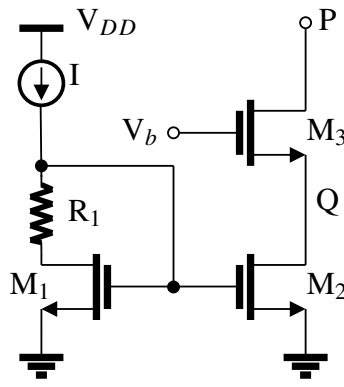


Figure 3: Low voltage cascode current mirror

$$V_{GS1} - IR_1 \geq V_{GS1} - V_{th1}$$

$$IR_1 \leq V_{th1}$$

But for minimum voltage headroom at P, $V_{DS2} = V_{GS2} - V_{th2} = V_{GS1} - V_{th2}$. Assuming $V_{th1} = V_{th2}$

$$V_{DS1} = V_{DS2} = V_{GS1} - V_{th1}$$

Therefore

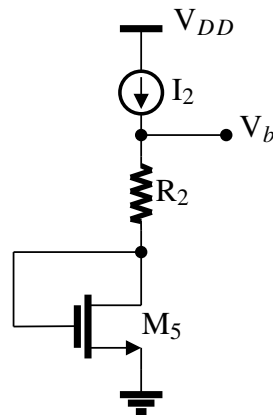
$$IR_1 = V_{th1}$$

Difficult to maintain $IR_1 = V_{th1}$ with PVT variations.

Generating V_b for above circuit for minimum voltage headroom at point P :

$$V_b = V_{GS3} + V_{GS2} - V_{th2} = V_{GS3} + V_{GS1} - V_{th1}$$

Option 1

Figure 4: Generation of V_b for low voltage cascode current mirror

If $V_{GS5} = V_{GS3}$ and $I_2 R_2 = V_{GS1} - V_{th1}$ then

$$V_b = V_{GS3} + V_{GS2} - V_{th2} = V_{GS3} + V_{GS1} - V_{th1} = V_{GS5} + I_2 R_2$$

$$V_{GS1} - V_{th1} = I_2 R_2$$

$$V_{GS1} - I R_1 = I_2 R_2$$

$$V_{GS1} = I R_1 + I_2 R_2$$

This is difficult to maintain as IR products do not track gate-source voltage. Following circuit can be used to tackle this issue.

Option 2

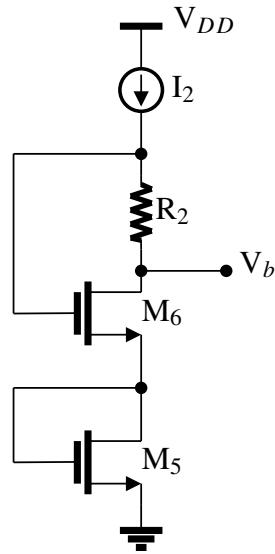


Figure 5: Generation of V_b for low voltage cascode current mirror

Here

$$V_b = V_{GS3} + V_{GS2} - V_{th2} = V_{GS3} + V_{GS1} - V_{th1} = V_{GS5} + V_{GS6} - I_2 R_2$$

For Figure 3

$$V_{DS1} = V_{GS1} - I R_1$$

$$V_{DS2} = V_b - V_{GS3}$$

For exact current copying, $V_{DS1} = V_{DS2}$

$$V_{GS1} - I R_1 = V_b - V_{GS3} = V_{GS5} + V_{GS6} - I_2 R_2 - V_{GS3}$$

But $V_{GS5} = V_{GS3}$

Therefore

$$V_{GS1} - I R_1 = V_{GS6} - I_2 R_2$$

Now V_{GS1} and V_{GS6} can track each other and IR_1 , $I_2 R_2$ track each other. There will be small error in copying

due to body effect.

To handle tracking of V_{th1} and IR_1 :

The level shift between gate and drain (as shown in Figure 3) need not be created by resistor. Here in Figure 6,

$V_{DS1} = V_b - V_{GS0}$. V_b can be chosen to place M_1 at the edge of saturation. $V_{DS1} = V_{DS2}$ if $V_{GS0} = V_{GS3}$.

We should choose V_b such that both M_0 and M_1 are in saturation.

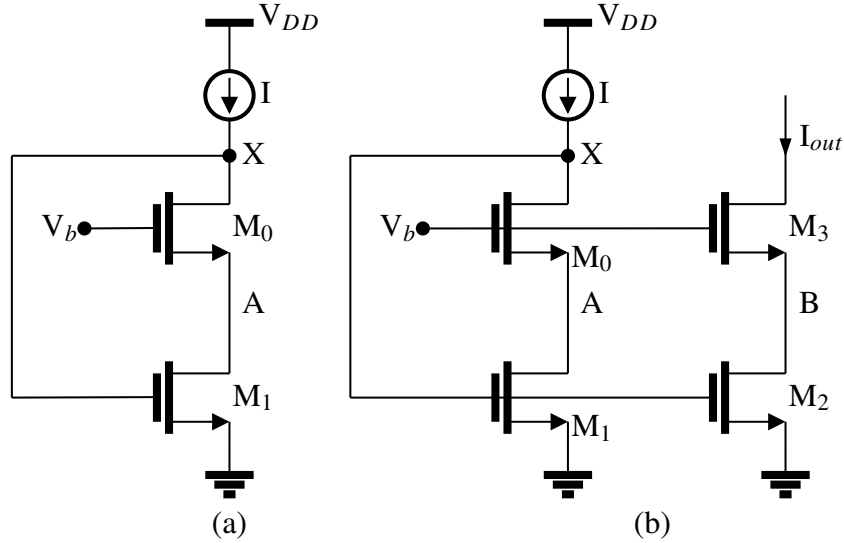


Figure 6: Modified low voltage cascode current mirror

For M_0 to be in saturation

$$V_X = V_{GS1} \geq V_b - V_{th0}$$

so

$$V_b \leq V_{GS1} + V_{th0}$$

For M_1 to be in saturation

$$V_A = V_b - V_{GS0} \geq V_{GS1} - V_{th1}$$

so

$$V_b \geq V_{GS0} + V_{GS1} - V_{th1}$$

Therefore

$$V_{GS0} + V_{GS1} - V_{th1} \leq V_b \leq V_{GS1} + V_{th0}$$

This is possible if

$$V_{GS0} + V_{GS1} - V_{th1} < V_{GS1} + V_{th0}$$

that is

$$V_{GS0} - V_{th1} < V_{th0}$$

or

$$V_{GS0} - V_{th0} < V_{th1}$$

We should size M_0 such that its overdrive is well below V_{th1} .

To generate V_b for circuit shown in Figure 6:

For minimum voltage headroom, $V_A = V_{GS1} - V_{th1}$ and hence V_b must be equal to (or slightly greater than) $V_{GS0} + V_{GS1} - V_{th1}$. The circuit is shown in Figure 7. M_5 generates $V_{GS5} \approx V_{GS0}$ and M_6 together with R_b

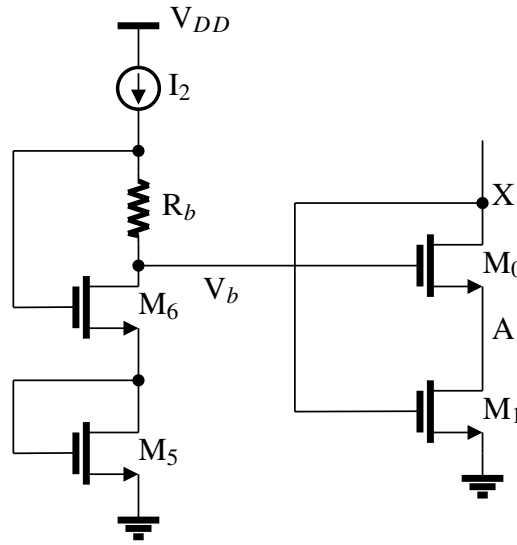


Figure 7: Generation of V_b for modified low voltage cascode current mirror

produces $V_{DS6} = V_{GS6} - I_2 R_b \approx V_{GS1} - V_{th1}$. Some inaccuracy nevertheless arises because M_5 does not suffer from body effect whereas M_0 does. Also, the magnitude of $I_2 R_b$ is not well-controlled.

Modified alternative to produce V_b is given below

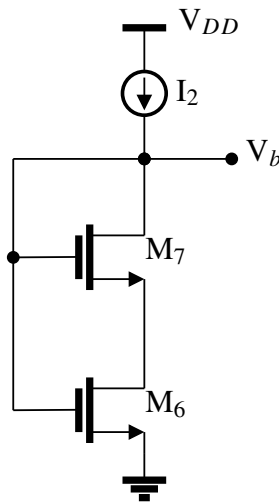


Figure 8: Generation of V_b for modified low voltage cascode current mirror

M_7 is always in saturation once ON. For M_6 to check region of operation, $V_{DS6} = V_{GS6} - V_{GS7}$. For M_6 to be in saturation, $V_{DS6} \geq V_{GS6} - V_{th6}$. Thus $V_{GS6} - V_{GS7} \geq V_{GS6} - V_{th6}$ or $V_{GS7} \leq V_{th6}$. This is not possible once M_7 is ON as $V_{th7} > V_{th6}$ due to body effect. Hence M_6 is in triode region once M_7 is ON.

Here

$$V_b = V_{GS7} + V_{DS6} = V_{GS6}$$

So we should design circuit such that $V_{GS7} = V_{GS0}$ and $V_{DS6} = V_{GS1} - V_{th1}$. But remember that M_6 is in triode region while M_1 is in saturation.

Currents flowing through M_6 and M_7 are same. M_6 is in triode while M_7 is in saturation. Therefore

$$\frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_7 V_{ov7}^2 = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_6 [2(V_{GS6} - V_{th6})V_{DS6} - (V_{DS6})^2]$$

But $V_{GS6} = V_{GS7} + V_{DS6}$ and $V_{DS6} = V_{GS1} - V_{th1}$, $V_{GS7} = V_{GS0}$. We have $V_{ov7} = V_{ov}$, $V_{DS6} = V_{GS1} - V_{th1} = V_{ov}$,
 $V_{GS6} = V_{GS7} + V_{DS6} = V_{th} + V_{ov} + V_{ov} = V_{th} + 2V_{ov}$

Therefore

$$\frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_7 V_{ov}^2 = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_6 [2(2V_{ov})V_{ov} - (V_{ov})^2]$$

and hence

$$\left(\frac{W}{L}\right)_6 = \frac{1}{3} \left(\frac{W}{L}\right)_7$$

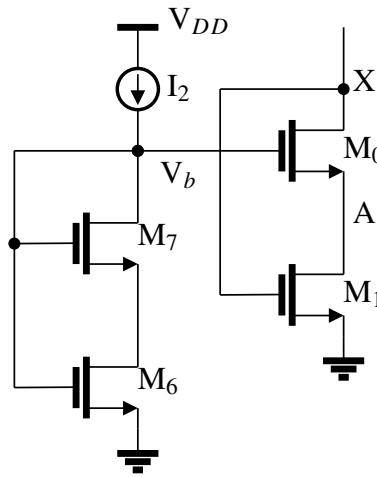


Figure 9: Generation of V_b for modified low voltage cascode current mirror

The overall circuit is given below.

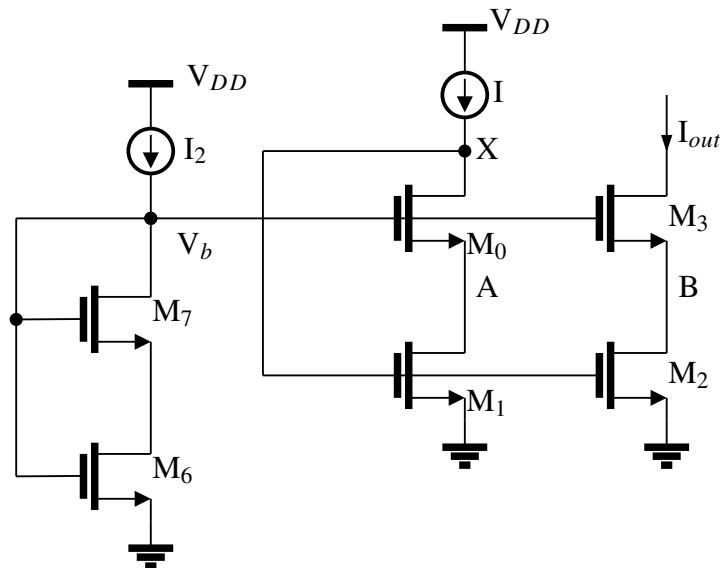


Figure 10: Modified low voltage cascode current mirror

If we tap the voltage from the drain of M_6 , it will be V_{ov} . For minimum headroom, voltage at the gate of M_1 should be $V_{ov} + V_{th}$ (which is already there) and voltage at the gate of M_3 should be $2V_{ov} + V_{th}$. So if we

add V_{ov} to the voltage $V_X = V_{ov} + V_{th}$, we can get this value. The modified diagram is given below.

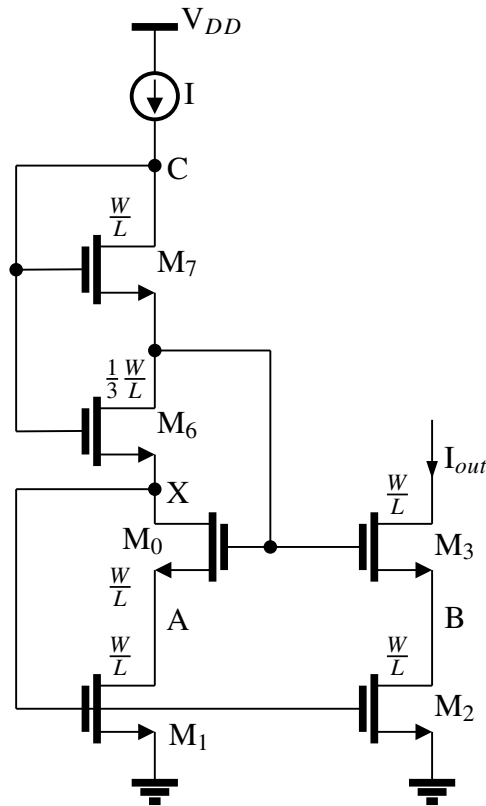


Figure 11: Modified low voltage cascode current mirror with lower power dissipation

The advantage of this is the lower power dissipation as we use single current source. The disadvantage is voltage requirement on input side of current mirror. At point C, minimum voltage is $3V_{ov} + 2V_{th}$.